

A Game-Theoretic Approach to Credit Card Fraud Detection

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Abstract. Intrusion prevention mechanisms are largely insufficient for protection of databases against Information Warfare attacks by authorized users and has drawn interest towards intrusion detection. We visualize the conflicting motives between an attacker and a detection system as a multi-stage game between two players, each trying to maximize his payoff. We consider the specific application of credit card fraud detection and propose a fraud detection system based on a game-theoretic approach. Not only is this approach novel in the domain of Information Warfare, but also it improvises over existing rule-based systems by predicting the next move of the fraudster and learning at each step.

1 Introduction

Intrusion detection is a critical part of the measures implemented for maintaining an attack tolerant database system. Though database management systems can provide intrusion prevention up to a certain extent by virtue of traditional access control mechanisms, they would not be sufficient for protection against syntactically correct but semantically damaging transactions [1]. Chung et al bring out that misuse detection in database systems has not been adequately addressed and propose DEMIDS, which can derive user profiles from database audit logs [2]. Lee et al suggest tagging the data objects with "time semantics" and monitor behavior at the level of sensor transactions [3]. Hu and Panda concentrate on analyzing the dependencies among data items in a database [4].

Consider a database system in an organization and a set of authorized users who have access rights on the database such as in banking services, credit card companies, etc. There always exists the possibility of legitimate and even non-legitimate transactions, what we hereby term as *fraudulent transactions*, being attempted by the authorized users or more typically, by adversaries posing as authorized users. The primary objective of any defense mechanism monitoring such an application would be to identify these fraudulent transactions as early as possible while limiting the possibility of raising too many false alarms. This form of Intrusion Detection in databases is an essential component of Information Warfare. The situation can be visualized as two adversaries playing against each other, the attacker launching attacks against the database system and the

detection system countering it. The problem effectively models as a typical game with each player trying to outdo the other and Game theory has long been used to tackle such problems.

The field of Game theory has been explored for problems ranging from auctions to chess and its application to the domain of Information Warfare seems promising. Samuel et al bring out the role of Game theory in Information Warfare [5]. They highlight that one can utilize well-developed Game theory algorithms to predict future attacks and the differences and challenges in this domain as compared to traditional games like chess, such as limited examples, multiple simultaneous moves and no time constraints [6]. Liu and Li have presented a game-theoretic attack prediction model for attacks on IDS-protected systems [7]. The authors have considered choosing a threshold by the detection system dependent on the profile of the customer and the availability weight provided by the system. It is clear that if the threshold is low, it may result in the genuine transactions being rejected causing a negative payoff to the cardholder, which is considered as zero by the authors. Further, in a real-world scenario, the genuine cardholder cannot be expected to choose his action according to Nash Equilibrium and any deviations can only be suspected. As the detection system increases the availability weight, to avoid denial of service to the customer, there is no pure strategy equilibrium and the thief can act to maximize his payoff. Our model is not limited by these assumptions and we also improvise by including the possibility of 'learning' in our system, which takes place at every step of the multi-stage game. This was also validated by our experimental study.

The rest of the paper is organized as follows. Section 2 describes the work related to credit card fraud detection. Section 3 describes the Game-theoretic model and the architecture of the proposed fraud detection system (FDS). In Section 4, we describe the experiment conducted and analyze the results. We conclude in Section 5 of the paper.

2 Related Work

Fraud, as in the Encyclopedia, is defined as "willful misrepresentation intended to deprive another of some right" and is a major source of concern in a number of applications such as e-commerce, telecommunication industry, computer intrusion, etc. Credit card fraud is a growing problem in the credit card industry. In the USA, the online retail sales were reported to be \$ 144 billion in 2004, which was a 26% increase over 2003 [8]. It is also estimated that 87% of purchases made over the Internet are paid by credit card [9]. The Association of Payment and Clearing Services (APACS) report showed that the cost of credit card fraud reached \$ 966.74 million in 2004, which was an increase of 20% as compared to 2003 [10]. Another survey of over 160 companies revealed that online fraud (committed over the Web or phone shopping) is 12 times higher than offline fraud (committed by using a stolen physical card) [11]. The growing number of credit card users worldwide provides more opportuni-